

Hierarchical Software-Defined Control for coordinated RAN and PON-based Transport Scaling

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Abstract—This demonstration shows the effectiveness of a hierarchical Software-Defined approach in coordinating Radio Access Network (RAN) and Passive Optical Network (PON)-based RAN transport to proactively scale virtual Distributed Units (vDUs) / Radio Units (RUs) and to jointly reconfigure the mid-haul transport.

Index Terms—Radio Access Networks, Software-Defined Networks, Passive Optical Networks, Hierarchical, Scaling

I. OVERVIEW

Network softwarization and virtualization unveil unprecedented potentials towards novel levels of efficiency and scalability. However, exploitation of these potentials is accompanied by novel challenges in terms of network control and management, especially in the access segment. Radio Access Networks (RAN) see the emergence of Open RAN paradigm (e.g., the one proposed by O-RAN) which combines the elasticity of Virtualized RANs (vRAN) and the controllability of Software Defined Networking (SDN) [1]. Parallely, passive optical networks (PONs) due to their increasing pervasiveness have gained attention as supporting infrastructures for mobile networks. The integration of RAN and PON via SDN has been shown to bring advantages in terms of network efficiency and total cost of ownership, thanks to the ability to dynamically adapt to varying network conditions and heterogeneous network requirements [2].

A critical aspect in the desired integration between RANs and PONs is represented by the latency introduced by multiple access procedures in PONs, which is particularly relevant in PON front-hauling and mid-hauling scenarios [3]. This becomes even more critical when PONs are used as mid-haul

transport in the cloud-native O-RAN, where virtual Distributed Unit (vDU) and virtual Central Unit (vCU) scaling call for dynamic activation and deactivation of dedicated resources in the interconnecting PON segment. Although the ability to efficiently adapt the vRAN workload to user traffic fluctuations via vDU/vCU scaling has been demonstrated, the realization of the end-to-end control and orchestration across all the involved domains still remains an open challenge.

In this demo, a hierarchical software-defined architecture is shown to be effective in proactively instantiating new vDUs and RUs in the RAN segment while an optical resource reconfiguration is applied in the PON-based mid-haul. A forecasting algorithm triggers the overall changes in the network by monitoring each vDU global upcoming traffic, as the cumulative maximum RUs capacity for a single vDU is approaching. This end-to-end (i.e., RU-DU-CU) configuration relies entirely on a transparent approach, where well-established components (i.e., the near Real-Time RIC (Near-RT RIC) and SDN Transport controller) are extended via app/xApps to provide a coordinated behavior.

II. INNOVATION

This paper presents several innovations from an end-to-end management perspective in the integration of RAN-Transport. The lack of centralized management in heterogeneous domains is a huge problem, which becomes evident in 5G and the upcoming 6G. Such networks were born with the idea of satisfying complex, low-latency, and high-demand scenarios. Their high degree of disaggregation is the enabling concept that makes them so powerful and flexible. On the other hand, they are inevitably dependent on the transport network that interconnects their disaggregated functions.

Passive Optical Networks are one of the proposed mid-haul (i.e., the interface between DU and CU) transport solutions in mobile networks, but require "awareness" of the context into which they are used. This can be achieved by performing

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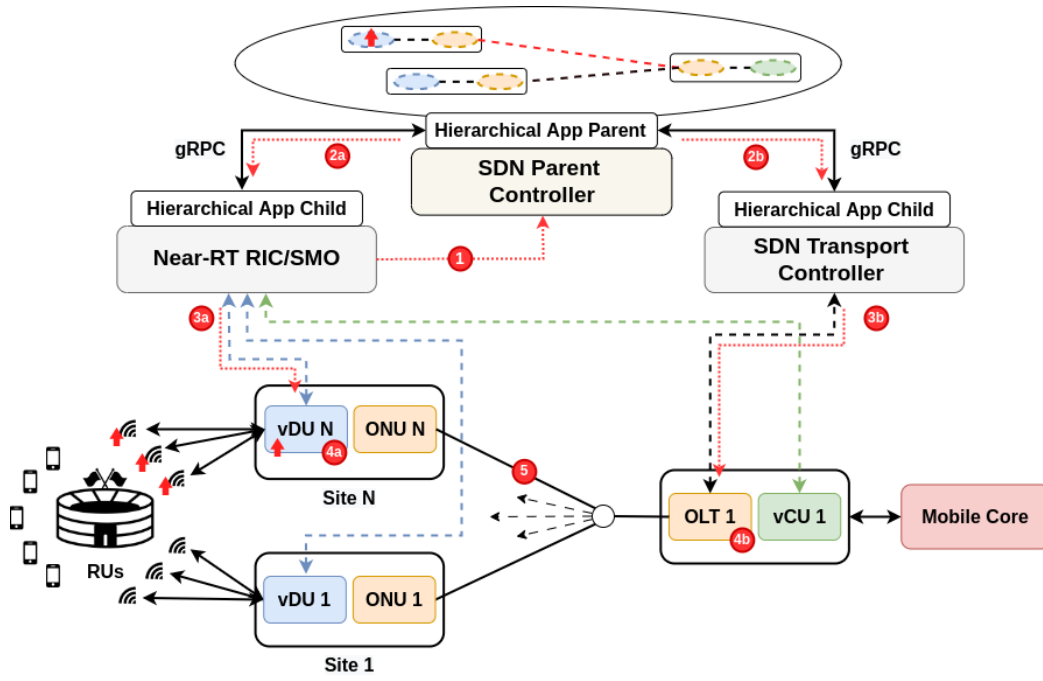


Fig. 1. Proposed demo scenario using a hierarchical SD solution to coordinate RAN and Transport networks.

strict communication of the RAN decisions to the PON's SDN controller, so that it can optimize and react to them. From the RAN perspective, scaling the number of DUs has several advantages, such as increased capacity during peak hours, smoother intra-gNB handovers, higher load balancing, and better resource management [4]. All of these scenarios find related actions in the PON, which require contextual awareness to maximise their impact.

A hierarchical solution achieves this type of coordination while still preserving a separate view of each domain. A hierarchical SDN controller enables high-level control over the different children by exploiting the imported global view. In this way, it is able to fine-tune decisions while allowing a certain degree of freedom for each child to determine how to best apply them in their own domain. By adopting the Near-RT RIC as a child SDN controller of the RAN and the PON SDN controller as a child SDN Controller of the Transport, an end-to-end orchestration framework can be achieved. The parent has visibility of the entire topology of its children, and it is also capable of implementing custom, application-defined logic as well as configuring intents concerning both domains.

The developed system leverages an innovative topology view synchronization model presented in [5], which guarantees resilient event propagation from child to parent controllers. This system has been extended to support the RAN environment, and also to forward decisions from parent to child.

The efficiency of the system is then enhanced by the use of a forecasting technique at the near-RT RIC/SMO. Using a Long-Short Term Memory (LSTM)-based algorithm, the child controller is able to proactively trigger reconfiguration at the parent according to the expected traffic at the vDUs. As a result, the coordinated behaviour offered by the parent

controller demonstrates its true effectiveness in terms of time when compared to a reactive approach.

Therefore, the proposed architecture provides a novel way to incorporate Passive Optical Networks into mobile networks, offering high-level proactive intent programmability spanning from radio to transport networks.

III. DEMO CONTENT & IMPLEMENTATION SECTION

Fig. 1 shows the architecture of the proposed solution. The scenario involves a central office that features a virtualized central unit (vCU1), bridging communication between the mobile core and remote sites where one or more vDUs are deployed, alongside the associated RUs.

The vDUs handle the low latency network functionalities (e.g., scheduling) and are connected to the radio units (RUs), implementing RF and part of the mobile Physical Layer functions. A Passive Optical Network is leveraged as mid-haul transport network between the vCU1, connected to the Optical Line Terminal (OLT), and the vDUs, deployed at the remote sites and connected to the Optical Network Unit (ONU).

From a management viewpoint, the PON is handled by the SDN Transport Controller, which is capable of monitoring and configuring optical resources using the NETCONF protocol. On the other hand, the Near-RT RIC manages the virtualized RAN and provides insights to the Service Management and Orchestration (SMO), which can scale in/out the components allocated over the network. Both the Near RT-RIC and the Transport controller are equipped with their relative child xApp/App, which forwards topology events using low-latency gRPC channels to a parent SDN controller. Therefore, the parent has visibility of all the information from the available

central/remote sites. This includes both OLTs and ONUs, but also the attached vDUs and vCUs in the network.

The actual implementation uses a containerized version of the different 5G RAN components, developed by OpenAirInterface (OAI) [6]. While the scenario provides for the scalability of vDUs based on the demand for more RUs, OAI does not yet support the usage of multiple RUs per vDU. For this reason, the implementation considers a single RU per vDU, but the approach can be generalized to multiple RUs. The different sites are deployed on different remote servers, attached to a commercial Calix Axos E7-2 NG-PON2 OLT via different ONUs. The OLT developed by Calix is compatible with the NETCONF protocol. In this way, it is possible to enhance the automation of control-plane while providing a seamless reconfiguration of the data-plane.

Each controller in the system is executed on an individual, dedicated machine. A Python-based controller is used for the Transport child, on which the Hierarchical Child App is installed. For the RAN, as long as no RIC is embedded with OAI by default, a Python script is used to emulate both the Near-RT RIC and the SMO behavior, as well as the child xApp. The parent controller is implemented using the open-source ONOS SDN controller [7]. On top of it, the Hierarchical Parent App is installed, implementing both the synchronization mechanism and the cross-domain intent logic for the demo use case. The ONOS GUI is used as the main graphical contribution of the demo, providing a reactive visualization of the different elements imported and configured on the child domains. The demonstration covers the following steps, depicted in Fig. 1:

1. The near-RT RIC/SMO monitors and forecasts the traffic handled by the vDU1, in the Remote Site 1, produced by the connected UEs/RUs. Once the forecasting algorithm foresees an increment of traffic which exceeds the 80% of the maximum capacity of the attached RUs on that site, the near-RT RIC/SMO produces a reconfiguration request towards the parent one to allocate new resources.
2. The parent controller analyzes the information available in the end-to-end topology, which was built from the messages sent by its child controllers. It then processes and generates a dedicated configuration request to the controllers handling those network segments: (2a) the Near-RT RIC/SMO and (2b) the SDN Transport controller.
- 3a. The Near-RT RIC/SMO, which manages different sites, instantiates the new vDU and activates the new RUs in Remote Site N towards vCU1.
- 3b. The SDN Transport controller, by processing the configuration embedded in the request, configures the OLT N towards ONU N, activating the connectivity among the two sites.
4. Finally, the inter-domain connectivity is established and new resources are allocated for the upcoming traffic. A virtualized UE is then attached to Remote Site N to generate traffic towards the Mobile core, showing the correctness of the end-to-end configuration.

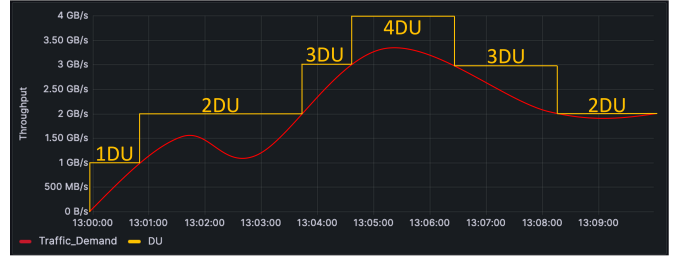


Fig. 2. Grafana panel showing the traffic demand along with the vDUs/RUs allocations.

5. The ONOS GUI, which receives topology updates from the children, displays the updated topology view, including the new vDU, and the PON link between Remote Site N and the OLT.

In the same way, additional remote sites can be activated configuring new vDUs, RUs and related path from the respective ONUs towards the OLT. The effectiveness of the system is also verified through a Grafana panel, which displays the incoming traffic demand and the relative DUs allocation as shown in Fig. 2.

Additional panels are then used to show the actual traffic passing through the different components after being re/configured.

IV. CONCLUSIONS

This demo showcased a hierarchical Software-Defined solution, capable of providing end-to-end configuration for a 5G/6G mobile network supported by a Passive Optical Network mid-haul. It enables an inter-domain, close interaction between the SDN Transport controller and the Near-RT RIC/SMO, allowing rapid and proactive scaling of virtual DUs/RUs with related resource provisioning along the optical path supporting the mid-haul.

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